Learning-based robotic systems rely on learned models and policies to operate in complex environments, enabling robot arms to manipulate household objects following language instructions, ground vehicles to navigate uneven terrain, and legged robots to climb stairs. However, performance deteriorates when deployment conditions differ from training: a manipulator may struggle when familiar objects are rearranged or viewed from a different camera angle, while a ground robot moving from concrete onto mud must adapt to changed traction. Such shifts can lead to actions inappropriate for the current environment, while collecting additional data and retraining for each new setting can be costly and may be impractical in remote or resource-constrained environments. **My research goal is to develop lightweight, self-supervised adaptation methods that enable robots to learn efficiently from deployment interactions, improve performance in unfamiliar environments, and support self-improvement and reliable real-world operation.** My undergraduate research experiences in Test-Time Adaptation (TTA) and Vision-Language-Action (VLA) models have shaped this goal and prepared me to pursue it. Through doctoral training, I hope to deepen my theoretical foundation, gain hands-on experience with physical robots, and refine my research judgment to identify and pursue original research questions.

Research Experience

ODE-Based Representation Learning. I have worked with Prof. Tao He since my second year at Sichuan University on TTA and representation learning with Ordinary Differential Equations (ODEs). My research began with ODE-based medical image segmentation, where ODEs model feature evolution during decoding. While discretized ODE models used low-order solvers for efficiency at the cost of accuracy, the fully continuous baseline used a high-order solver but incurred substantial computational and memory costs. To address this trade-off, I developed CNM-UNet, combining a high-order ODE solver with a single decoder that replaces vanilla UNet's hierarchical decoders, retaining the accuracy of continuous modeling while reducing computational overhead. Cross-domain experiments also revealed their vulnerability to distribution shift, so I designed DUSE, a TTA strategy that updates selected decoder parameters and learns image prompts from unlabeled test images. On ISIC2018 and three polyp datasets, CNM-UNet achieved competitive accuracy at lower computational cost than the fully continuous baseline. DUSE further improved cross-domain performance and robustness to image corruptions. ***This project resulted in a first-author paper at AAAI 2026 [1], and I publicly released its code.*** As my first research project, it taught me to communicate effectively with my advisor and gave me confidence to pursue further research. More importantly, it changed how I judge model reliability: strong performance on test data resembling the training distribution does not guarantee robustness under larger shifts, especially in real-world deployment. This lesson sparked my interest in generalization and led me to further work on TTA.

Test-Time Adaptation. Following CNM-UNet, I explored parameter-selective TTA for medical image segmentation under Prof. Tao He. Parameter-selective TTA restricts adaptation to a subset of parameters to limit harmful updates and preserve source-domain knowledge. However, existing methods often treat layers independently, overlooking feature dependencies, and noisy self-supervision can destabilize updates during continual adaptation. To address these problems, I developed SaCTTA. I designed an attention-driven layer-selection mechanism that uses cross-layer dependencies to select and fuse useful layer features, and an adaptive learning-rate strategy that combines parameter sensitivity with divergence from source attention patterns to regulate update magnitude and mitigate error accumulation. Across optic disc/cup, polyp, and prostate MRI segmentation, SaCTTA achieved the highest average performance among the compared adaptation methods, with additional evaluations supporting its effectiveness with convolutional and transformer backbones. ***This work resulted in a co-first-authored manuscript under review at Neurocomputing [2].*** This project showed me that even strong parameter-selective methods can degrade under long, continually changing test streams, leading me to ask whether a general mechanism could mitigate the effects of non-stationary distribution shifts.

From February to September 2026, working with Professor Tao He, I examined the influence of non-stationary test distributions on TTA methods. Existing TTA methods mainly regulate objectives, samples, or model parameters, but rarely examine whether accumulated optimizer history remains compatible with the current test distribution. Most rely on stateful optimizers that retain update directions from earlier test batches. I suspected that directions accumulated in one domain might continue to influence updates after a shift, even when they no longer matched the current data. To test this hypothesis, I reset only optimizer states at known domain boundaries while keeping adapted parameters unchanged. Error decreased across all five optimizers, supporting my hypothesis. I subsequently developed P-DORM, a plug-and-play wrapper that compares current gradients with historical update directions and selectively resets incompatible optimizer states. I designed the reset criterion to account for how gradient alignment normally varies across parameters.

Across four datasets and six TTA methods, P-DORM improved average performance on diverse test streams with both convolutional and transformer backbones. ***This work resulted in a first-authored manuscript currently under review at ICLR 2027 [3].*** Although the test streams in this work were manually constructed, this project made me appreciate the challenge of continually changing test distributions and the importance of managing such changes for reliable real-world deployment.

Vision-Language-Action Models. During the poster sessions at AAAI 2026, I learned about VLA models and became particularly interested in how they combine visual observations and language instructions to guide robot actions. From March to October 2026, I worked remotely with Prof. Yuzhang Shang at the University of Central Florida on reasoning and generalization for VLA models. Reasoning-based VLAs often regenerate textual plans or visual subgoals before each action, raising inference cost. I explored whether a global plan generated once per task could provide persistent guidance while maintaining performance. Initially, I combined global planning and action generation within a single VLA. However, task-start samples (t = 0) were scarce, and oversampling them increased training cost while reducing the proportion of later-step samples per batch. This version achieved only 15% success. I therefore separated planning and action generation into a dual-system structure. I froze the upper VLM backbone and trained only query tokens and projection layers to learn global representations from limited task-start data; the lower policy generated actions from these representations, task instructions, and current observations throughout execution. On LIBERO, this system reached 81% average success, though below my expectations. Alongside this work, I proposed deriving rewards from the model's own reasoning and interaction history for self-supervised policy improvement. This project gave me hands-on experience in VLA training and evaluation and taught me to examine how action policies learn and use reasoning representations. The computational demands of training VLA policies from scratch, together with my TTA research, motivated me to explore lightweight methods that help pretrained policies learn efficiently from deployment interactions.

Research Interests and Future Directions

In the future, I hope to develop self-improving robotic models that learn efficiently from deployment experience and adapt reliably to unfamiliar environments. I am particularly interested in the following research directions.

Generating Reliable Training Data During Deployment. A common approach to improving real-world robot performance is to train on data collected in the target environments. However, scaling this approach across diverse environments can be costly: collecting demonstrations and annotating robot experience requires substantial human effort, including teleoperation and reviewing execution outcomes. During deployment, a robot takes actions and observes their consequences. For example, an attempted grasp may successfully lift an object, or the object may slip. These outcomes reveal how its actions affect the environment and could provide informative learning signals beyond human demonstrations. I am interested in developing self-supervised methods that turn these interactions into training data by deriving reliable supervision from action outcomes. Such methods could reduce the need for repeated human demonstrations and help policies learn from real-world conditions not represented in their original training data. A key challenge is determining which experiences provide useful supervision, so that learning improves task performance rather than reinforces the robot's mistakes.

Efficient Adaptation During Deployment. Pretrained robotic policies can benefit from further learning under unfamiliar deployment conditions, but full-model updates require substantial computation and memory, making frequent adaptation difficult on resource-constrained robots. Parameter-efficient methods such as LoRA reduce these costs, yet activation storage and gradient computation can still impose substantial overhead. I am interested in developing lightweight adaptation methods that let pretrained policies learn efficiently from deployment experience. One direction is to train lightweight residual modules while keeping the pretrained policy frozen. For black-box policies, an external action-correction module could adapt behavior without accessing the underlying weights or gradients. A key challenge is determining where to insert such modules and how much capacity to add, so that limited updates improve behavior without destabilizing previously learned skills or accumulating errors across deployments. My work on parameter-selective TTA provides a foundation for this research.

Learning from the Environment. Real-world deployment exposes robots to physical conditions that may differ from those encountered during training. For example, an object may be heavier than expected, or a grasp may slip because of a change in contact friction. Such differences affect how actions change the environment and can cause familiar behaviors to fail. I am interested in developing self-supervised methods that learn representations of deployment-specific physical conditions from a robot's interaction history and use them to inform action generation. Such methods could help policies adjust their behavior to unfamiliar dynamics without requiring explicit human annotations of physical

properties. This ability may be particularly valuable in sim-to-real transfer or when robots encounter conditions poorly represented in their training data, such as soft or deformable terrain, and could support more reliable operation in unfamiliar environments.

Safe Exploration During Deployment. Learning from deployment interactions allows robots to acquire new experience, but exploratory actions can damage objects or put nearby people at risk. Collision avoidance alone cannot address all such risks. For example, a collision-free motion could still place a hot container within a child's reach. I am interested in developing methods that enable robots to gather informative experience while respecting physical constraints and accounting for context-dependent hazards. In particular, I hope to explore how context-based safety judgments can be combined with physical constraints to guide exploratory action generation. Such methods could reduce risks to the robot and its surroundings while making self-improvement more practical in spaces shared with people.

Through doctoral research, I hope to connect my adaptation research with experiments on physical robots and apply mathematical tools to robot-learning problems. After completing my PhD, I hope to pursue a career as a research scientist in industry, advancing robots' capabilities in real-world applications.

References

Publications

[1] **Tianqi Xu**, Yashi Zhu, Quansong He, Yue Cao, Kaishen Wang, Zhang Yi, and Tao He. CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation. *AAAI 2026*.

Under Review

[2] **Tianqi Xu**[*], Jun Wu[*], Junjie Hu, Jixiang Guo, Wei Liu, Lei Zhang, Zhang Yi, and Tao He. SaCTTA: Selective Attention Continual Test-Time Adaptation for Medical Image Segmentation. *Neurocomputing*, under review. [*]Equal contribution.

[3] **Tianqi Xu**, Junjie Hu, Jixiang Guo, Haixian Zhang, Zhang Yi, and Tao He. P-DORM: Parameter-Wise Dynamic Optimizer-State Reset for Test-Time Adaptation. *ICLR 2027*, under review.